

ITCS 4114: Real World Algorithms

Math Refresher

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Mathematical Fundamentals

- ◇ Sets
- ◇ Factorials, Logarithms
- ◇ Recursion
- ◇ Summations, Recurrences
- ◇ Proof Techniques: By Contradiction, Induction
- ◇ Estimation Techniques

Reference: Levitin, Appendix A

Sets

A collection of distinguishable members or elements drawn from a base type

Set Notation: if **A** and **B** are sets:

- ◇ Example: $\mathbf{A} = \{1, 4, 7, 2\}$
- ◇ Positive integer Set: $\{x | x \text{ is a positive integer}\}$
- ◇ Membership: $x \in \mathbf{A}$, or, $x \notin \mathbf{A}$
- ◇ Empty Set: $\emptyset$
- ◇ Size: $|\mathbf{A}|$
- ◇ Subset, Superset: $\mathbf{A} \subseteq \mathbf{B}$, $\mathbf{B} \supseteq \mathbf{A}$
- ◇ Union: $\mathbf{A} \cup \mathbf{B}$ // elements in A and B
- ◇ Intersection: $\mathbf{A} \cap \mathbf{B}$ // elements common to A and B
- ◇ Difference: $\mathbf{A} - \mathbf{B}$ // elements present in A, not in B

- ◇ **Linear Ordering:** Any pair of set elements have a unique order, one of $<$, $>$, or $=$.
- ◇ **Transitivity:** $a < b$ and $b < c$ implies $a < c$.
- ◇ **Finite Sequence:** Sequence of length n drawn from $(0, 1, \dots, n-1)$, and have an ordering; may contain duplicates.

Relevance

Exhaustive Search Problems (Traveling Salesman, Assignment, Knapsack problems)

Miscellaneous Notation

- ◇ **Factorial** of n is defined as

$$n! = n(n-1)(n-2) \cdots (3)(2)(1), \text{ with } 0! = 1$$

- ◇ **Permutation:** Members of a sequence arranged in some order.
- ◇ **Floor** of x , $\lfloor x \rfloor$, returns the greatest integer not larger than x .
- ◇ **Ceiling** of x , $\lceil x \rceil$, returns smallest integer not smaller than x .
- ◇ **Modulus** returns remainder of integer division.

Functions (Computational Complexity Related)

- ◇ **Constant:** $f(n) = c$, where c is a constant.
- ◇ **Linear:** $f(n) = cn$, where c is a constant.
- ◇ **Quadratic:** $f(n) = cn^2$, where c is a constant.
- ◇ $f(n) = c n \log n$

Logarithms

Definition

- ◇ *Logarithm* of **base** b for **value** y is 'the **power** to which b must be raised to get y '

$$\log_b y = x \Leftrightarrow b^x = y \Leftrightarrow b^{\log_b y} = y$$

- ◇ **Applications:** Algorithm analysis (binary search, tree structures)

Properties: For positive values m, n, a, r

$$\log_a 1 = 0$$

$$\log_a a = 1$$

$$\log_a nm = \log_a n + \log_a m$$

$$\log_a n/m = \log_a n - \log_a m$$

$$\log_a n^r = r \log_a n$$

$$\log_a n = \frac{\log_b n}{\log_b a} = \log_b n \log_a b$$

Combinatorics

- ◇ Number of permutations of an n element set:

$$P(n) = n! = \prod_{i=1}^n i = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$$

- ◇ Number of combinations, $C(n, k)$ each of size k from an n element set: $C(n, k) = \frac{n!}{k!(n-k)!}$
- ◇ Number of subsets in an n -element set : 2^n

Recursion: An algorithm is **recursive** if it calls itself to do part of its work.

- ◇ Needs a **Base** case - simple input that can be solved without recursion.
- ◇ Recursive part - which leads towards the base case.
- ◇ **Examples:** Factorial, Fibonacci Sequence, Towers of Hanoi

Summations

General form:

$$\sum_{i=1}^n f(i) = f(1) + f(2) + \cdots + f(n)$$

Closed Form:

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n \frac{1}{2^i} = 1 - \frac{1}{2^n}$$

$$\sum_{i=1}^n 2^i = 2^{n+1} - 2$$

Recurrence Relations

- ◇ Useful in analyzing **recursive** algorithms (merge sort, quicksort).
- ◇ Recurrence relations defines a function by including smaller instances of the problem, such as

$$n! = n.(n-1)!, 1! = 0! = 1$$

$$Fib(n) = Fib(n-1) + Fib(n-2), Fib(1) = Fib(2) = 1$$

- ◇ Towers of Hanoi recurrence:

$$T(n) = 2T(n-1) + 1$$

Mathematical Proof Techniques

Proof by Contradiction

Assume the theorem is false.

Find a logical contradiction from this assumption.

If the logic used is correct, then the original assumption is wrong and the theorem is correct.

Example

There is no largest integer.

Proof.

Assume there is a largest integer, N .

Add 1 to N , resulting in $N' = N + 1$.

N' is an integer formed by adding two integers, and $N' > N$, contradicting the assumption.

Proof by Induction

To prove a theorem **T** is true, express **T** in terms of positive integer n , and then proceed similar to recursion:

Base case: Show **T** holds for $n = c$

Induction Step: Assume **T** holds for some $k < n$, and show it holds for $n = k + 1$

Induction Problems

Prove by induction the following:

- ◇ $\sum_{i=1}^n i = \frac{n(n+1)}{2}$
- ◇ $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$
- ◇ $\sum_{i=0}^n r^i = \frac{1-r^{n+1}}{1-r}$ (geometric series sum)
- ◇ $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$
- ◇ $\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$
- ◇ $\sum_{i=0}^n a^i = \frac{a^{n+1}-1}{a-1}$ for $a > 1$
- ◇ $\sum_{i=0}^n 2^i = 2^{n+1} - 1$
- ◇ $\sum_{i=1}^{n-1} 2i - 1 = (n-1)^2$
- ◇ $\sum_{i=0}^n i!i = (n+1)! - 1$
- ◇ $\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$

Contradiction Problems

Prove by contradiction the following problems:

- ◇ There is no largest integer.
- ◇ If n is an integer and n^2 is even, then n is even
- ◇ If k is an integer and k is odd, then k^2 is odd
- ◇ For all integers n , if $n^3 + 5$ is odd, then n is even
- ◇ $\sqrt{2}$ is irrational.
- ◇ If z is an integer and z^2 is odd, then z is odd.
- ◇ There do not exist two integers $m, n \in \mathbb{Z}$ such that $n^4 - 4m = 2$